

DataVisualization_hb

August 6, 2024

1 Further Data Visualization of the Dataset

```
[ ]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import calendar

# Load the overdose data
file_path = "/Users/hunterberberich/Desktop/Data/vis_df.csv"
df = pd.read_csv(file_path)
df['date'] = pd.to_datetime(df['date'])
df['month'] = df['date'].dt.month
df['year'] = df['date'].dt.year
```

```
[ ]: print(df.dtypes)
df
```

date	datetime64[ns]
tempmax	float64
tempmin	float64
temp	float64
feelslikemax	float64
feelslikemin	float64
feelslike	float64
dew	float64
humidity	float64
precip	float64
precipcover	float64
preciptype	object
snow	float64
snowdepth	float64
cloudcover	float64
uvindex	int64
severerisk	float64
day	object
gender_desc	object
race	object
susp_od_drug_desc	object

```

naloxone_administered    object
survive                   object
month                     int32
year                      int32
dtype: object

```

```

[ ]:      date  tempmax  tempmin  temp  feelslikemax  feelslikemin  \
0   2018-01-07    26.2    -4.2  10.1         18.1         -16.6
1   2018-01-09    35.8    28.0  32.5         34.7          24.1
2   2018-01-14    15.3     2.5   9.2         14.6          -6.0
3   2018-01-14    15.3     2.5   9.2         14.6          -6.0
4   2018-01-14    15.3     2.5   9.2         14.6          -6.0
...
3679 2024-04-11    68.8    58.9  61.6         68.8          58.9
3680 2024-04-12    59.4    46.8  50.8         59.4          39.8
3681 2024-04-13    59.2    44.5  51.1         59.2          37.4
3682 2024-04-14    81.1    45.0  64.1         79.9          39.9
3683 2024-04-14    81.1    45.0  64.1         79.9          39.9

```

```

      feelslike  dew  humidity  precip  ...  uvindex  severerisk  day  \
0           0.6 -3.8      57.2  0.000  ...      2         0.0  Sunday
1          26.8 26.6      78.9  0.000  ...      2         0.0  Tuesday
2           3.5  0.5      68.0  0.000  ...      2         0.0  Sunday
3           3.5  0.5      68.0  0.000  ...      2         0.0  Sunday
4           3.5  0.5      68.0  0.000  ...      2         0.0  Sunday
...
3679        61.6 56.7      84.8  1.526  ...      3        10.0  Thursday
3680        46.5 46.3      84.5  0.266  ...      2        10.0  Friday
3681        47.9 33.0      51.5  0.019  ...      9        10.0  Saturday
3682        62.9 39.5      42.6  0.001  ...      8        30.0  Sunday
3683        62.9 39.5      42.6  0.001  ...      8        30.0  Sunday

```

```

      gender_desc  race  susp_od_drug_desc  naloxone_administered  survive  \
0           Male  BLACK             HEROIN                      N        Y
1           Male  WHITE             HEROIN                      Y        Y
2           Male  WHITE             HEROIN                      Y        Y
3           Male  WHITE             HEROIN                      Y        Y
4           Male  WHITE             HEROIN                      Y        Y
...
3679        Female  BLACK             FENTANYL                  Y        U
3680        Female  WHITE             HEROIN                    Y        Y
3681           Male  WHITE             HEROIN                    Y        Y
3682           Male  WHITE             HEROIN                    Y        Y
3683        Female  BLACK             HEROIN                    Y        U

```

```

      month  year
0          1  2018

```

```

1      1  2018
2      1  2018
3      1  2018
4      1  2018
...    ...  ...
3679   4  2024
3680   4  2024
3681   4  2024
3682   4  2024
3683   4  2024

```

[3684 rows x 25 columns]

```
[ ]: num_columns = df.select_dtypes(include=['float64', 'int64']).columns
cat_columns = df.select_dtypes(include=['object']).columns
df2 = df.drop(columns=['date', 'month', 'year'])
```

```
[ ]: key_variables = ['tempmax', 'tempmin', 'feelslike', 'humidity', 'precip',
↳ 'cloudcover']
```

```
[ ]: five_number_summary2 = df[key_variables].describe().loc[['min', '25%', '50%',
↳ '75%', 'max']]
print(five_number_summary2)
```

	tempmax	tempmin	feelslike	humidity	precip	cloudcover
min	11.4	-5.000	-13.2	26.000	0.0000	0.000
25%	48.5	32.800	35.7	60.075	0.0000	30.175
50%	66.8	47.500	56.0	70.450	0.0050	55.900
75%	78.6	61.625	69.7	79.700	0.1105	80.700
max	93.7	75.500	87.2	99.200	3.9220	100.000

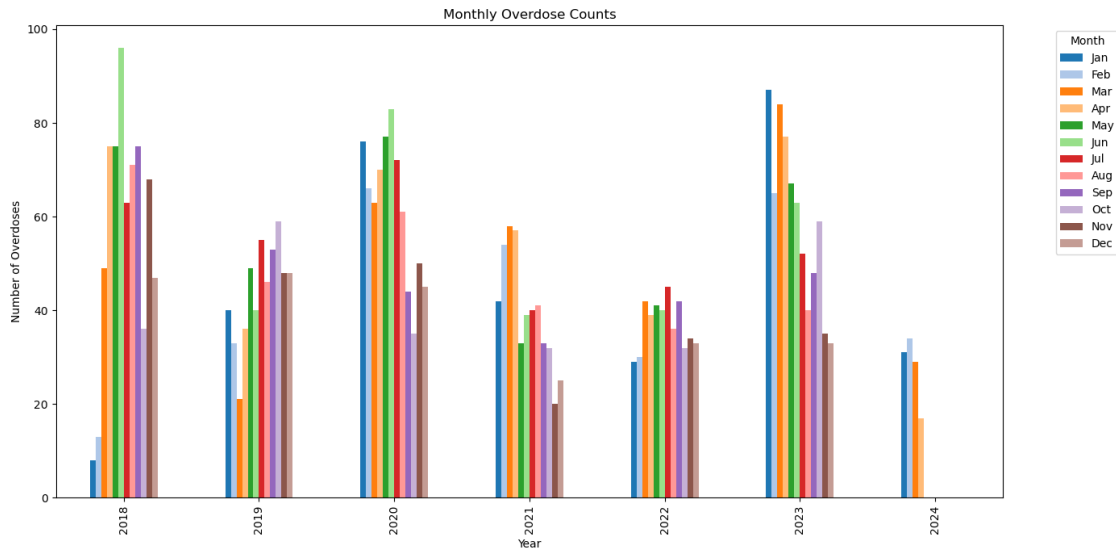
```
[ ]: monthly_counts = df.groupby(['year', 'month']).size().unstack(fill_value=0)
monthly_counts.columns = [calendar.month_abbr[m] for m in monthly_counts.
↳ columns]
```

```

plt.figure(figsize=(14, 7))
colors = plt.cm.tab20.colors # Using tab20 colormap
monthly_counts.plot(kind='bar', stacked=False, figsize=(14, 7), color=colors)
plt.title('Monthly Overdose Counts')
plt.xlabel('Year')
plt.ylabel('Number of Overdoses')
plt.legend(title='Month', bbox_to_anchor=(1.05, 1), loc='upper left')
plt.tight_layout()
plt.show()

```

<Figure size 1400x700 with 0 Axes>



```
[ ]: top_days = df['date'].value_counts().nlargest(3)

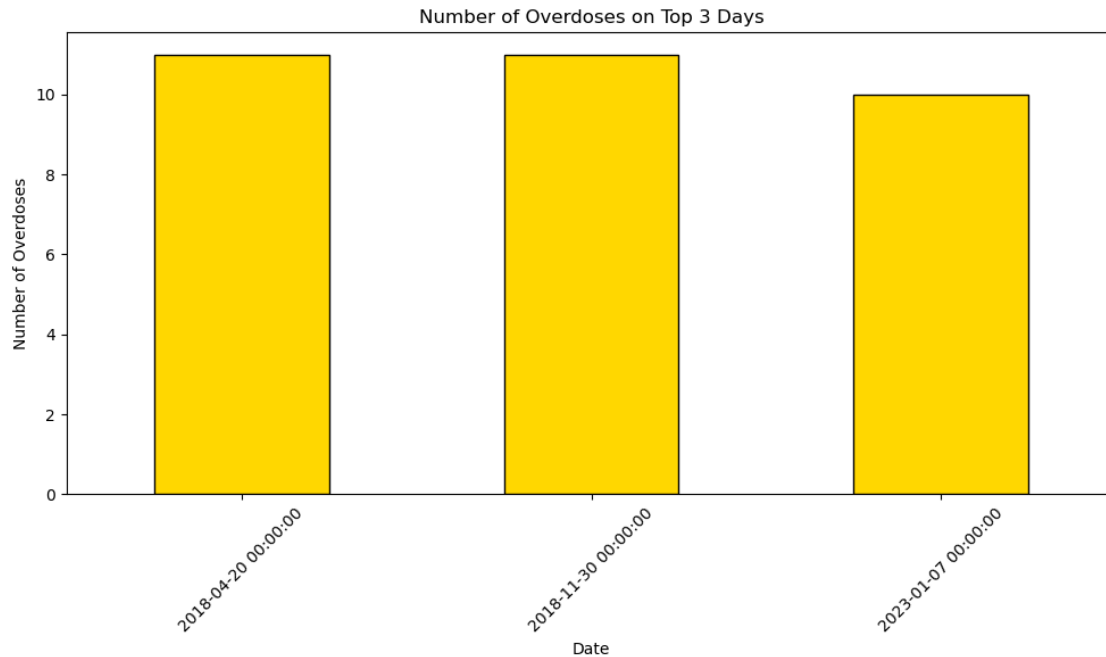
print(top_days)
```

```
date
2018-04-20    11
2018-11-30    11
2023-01-07    10
Name: count, dtype: int64
```

```
[ ]: top_days_weather = df[df['date'].isin(top_days.index)][['date', 'tempmax',
↳ 'tempmin', 'temp', 'feelslike', 'precip', 'snow',
↳ 'cloudcover', 'day', 'preciptype']]

# Display the weather conditions for these days
top_days_weather

plt.figure(figsize=(10, 6))
top_days.plot(kind='bar', color='gold', edgecolor='black')
plt.title('Number of Overdoses on Top 3 Days')
plt.xlabel('Date')
plt.ylabel('Number of Overdoses')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

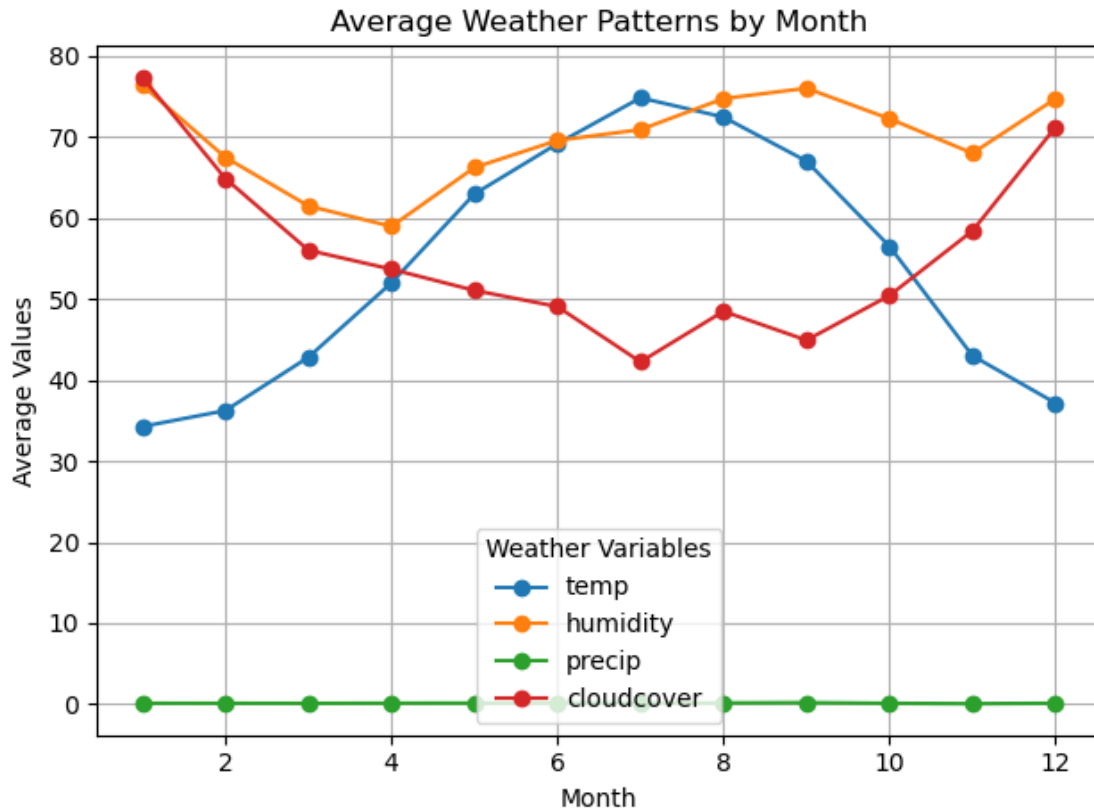


```
[ ]: #The three days with the highest amount of OD were all on Friday and Saturday,
      ↪ it was chilly, and 2/3 it was snowing.
```

```
[ ]: monthly_weather = df.groupby('month')[['temp', 'humidity', 'precip',
      ↪ 'cloudcover']].mean()
```

```
plt.figure(figsize=(14, 7))
monthly_weather.plot(kind='line', marker='o')
plt.title('Average Weather Patterns by Month')
plt.xlabel('Month')
plt.ylabel('Average Values')
plt.legend(title='Weather Variables')
plt.grid(True)
plt.tight_layout()
plt.show()
```

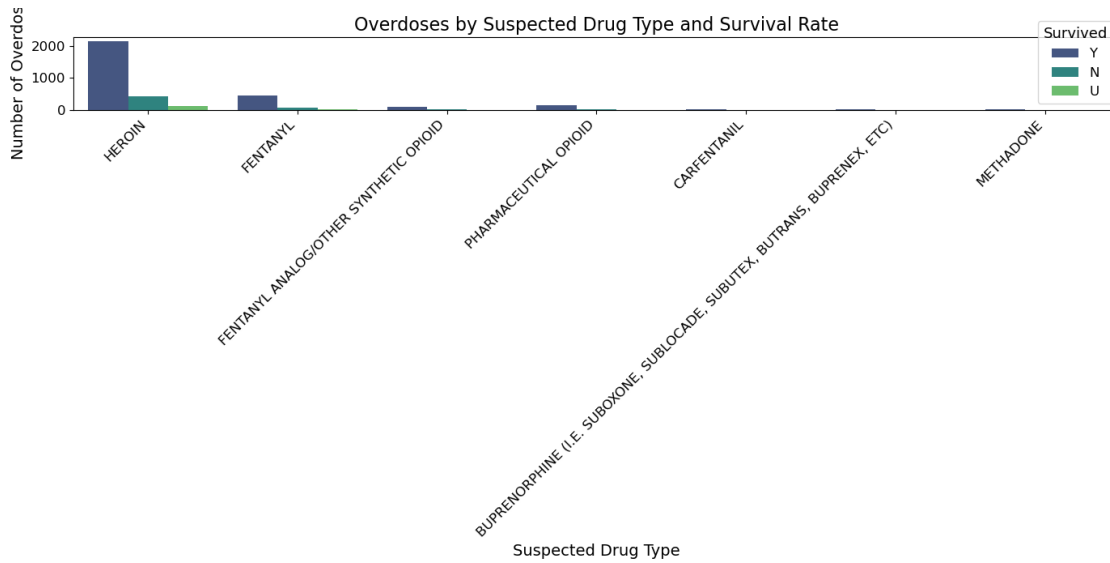
<Figure size 1400x700 with 0 Axes>



```
[ ]: plt.figure(figsize=(14, 7))
sns.countplot(data=df, x='susp_od_drug_desc', hue='survive', palette='viridis')

plt.title('Overdoses by Suspected Drug Type and Survival Rate', fontsize=16)
plt.xlabel('Suspected Drug Type', fontsize=14)
plt.ylabel('Number of Overdoses', fontsize=14)
plt.legend(title='Survived', title_fontsize='13', fontsize='12')
plt.xticks(rotation=45, ha='right', fontsize=12)
plt.yticks(fontsize=12)

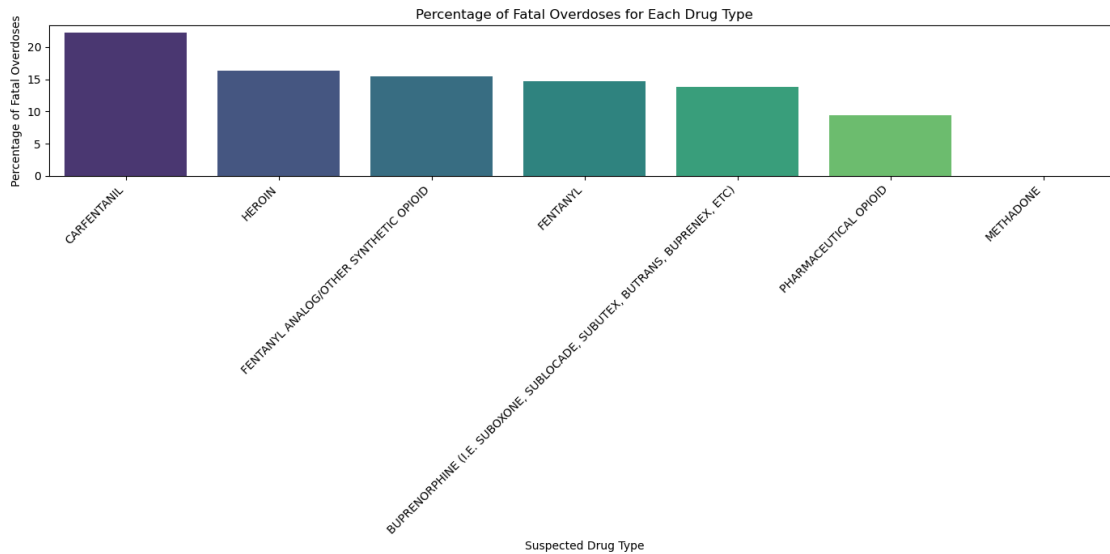
plt.tight_layout()
plt.show()
```



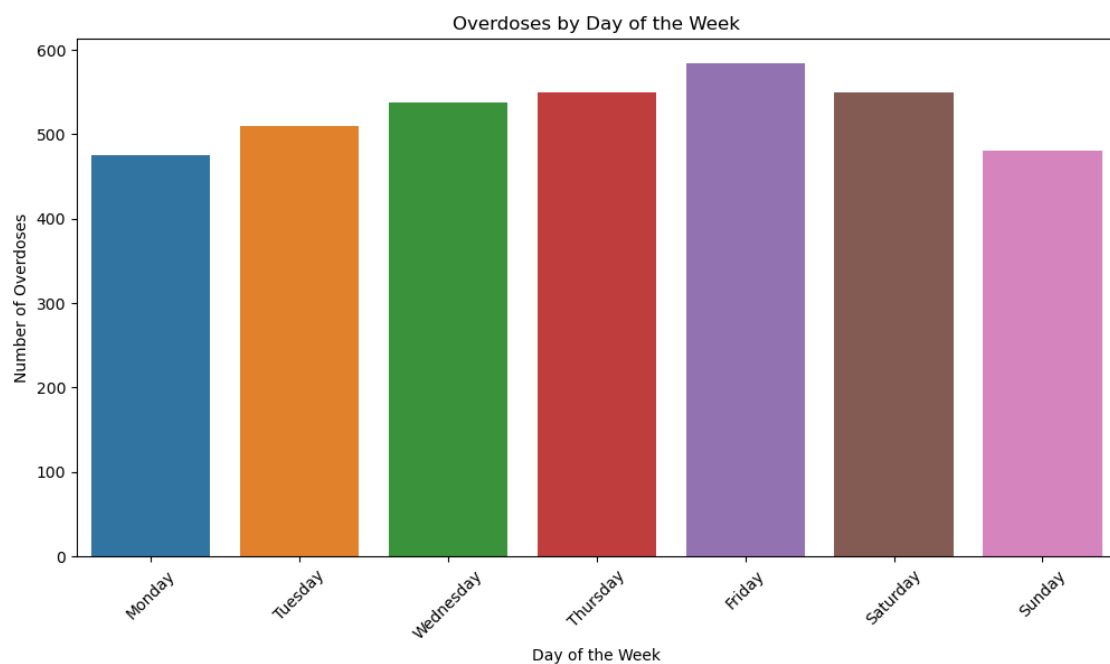
```
[ ]: survival_counts = df.groupby('susp_od_drug_desc')['survive'].value_counts().
    ↪unstack(fill_value=0)
survival_counts['Fatal Overdose Rate'] = survival_counts['N'] /
    ↪(survival_counts['Y'] + survival_counts['N']) * 100

survival_counts = survival_counts.sort_values(by='Fatal Overdose Rate',
    ↪ascending=False)

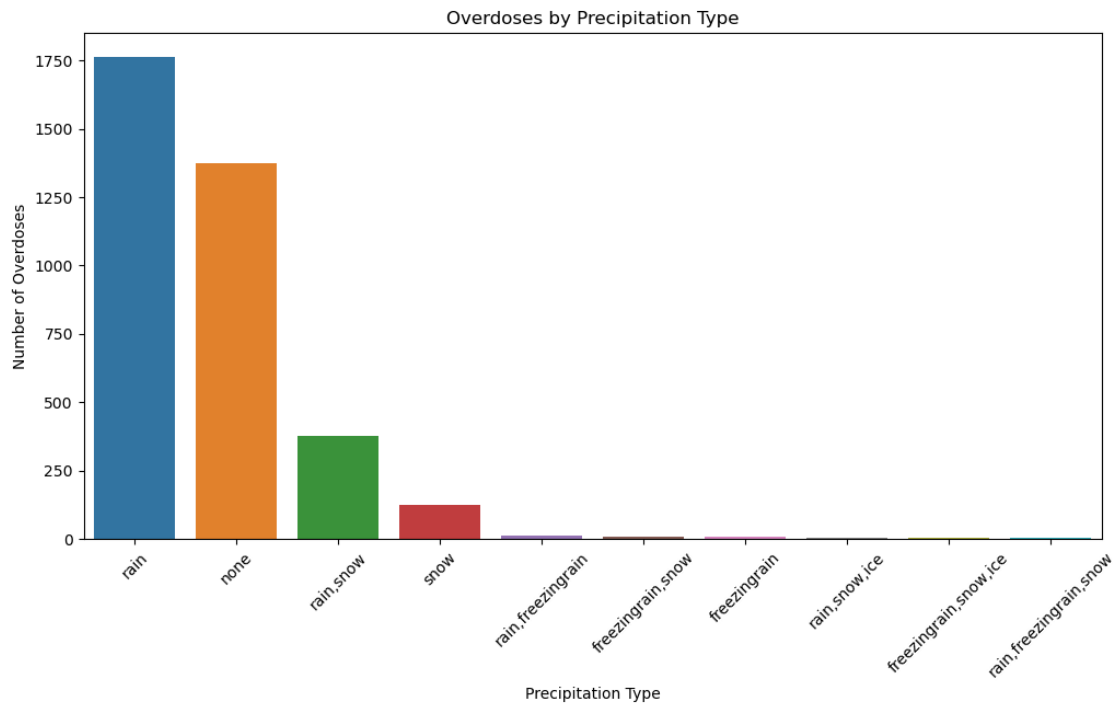
plt.figure(figsize=(14, 7))
sns.barplot(x=survival_counts.index, y=survival_counts['Fatal Overdose Rate'],
    ↪palette='viridis')
plt.title('Percentage of Fatal Overdoses for Each Drug Type')
plt.xlabel('Suspected Drug Type')
plt.ylabel('Percentage of Fatal Overdoses')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```



```
[ ]: plt.figure(figsize=(12, 6))
sns.countplot(data=df, x='day', order=['Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturday', 'Sunday'])
plt.title('Overdoses by Day of the Week')
plt.xlabel('Day of the Week')
plt.ylabel('Number of Overdoses')
plt.xticks(rotation=45)
plt.show()
```



```
[ ]: plt.figure(figsize=(12, 6))
sns.countplot(data=df, x='preciptype', order=df['preciptype'].value_counts().
    ↪index)
plt.title('Overdoses by Precipitation Type')
plt.xlabel('Precipitation Type')
plt.ylabel('Number of Overdoses')
plt.xticks(rotation=45)
plt.show()
```



```
[ ]: # Bar chart of overdoses by gender
plt.figure(figsize=(10, 5))
sns.countplot(data=df, x='gender_desc')
plt.title('Overdoses by Gender')
plt.xlabel('Gender')
plt.ylabel('Number of Overdoses')
plt.show()

# Bar chart of overdoses by race
plt.figure(figsize=(12, 6))
sns.countplot(data=df, x='race')
plt.title('Overdoses by Race')
```

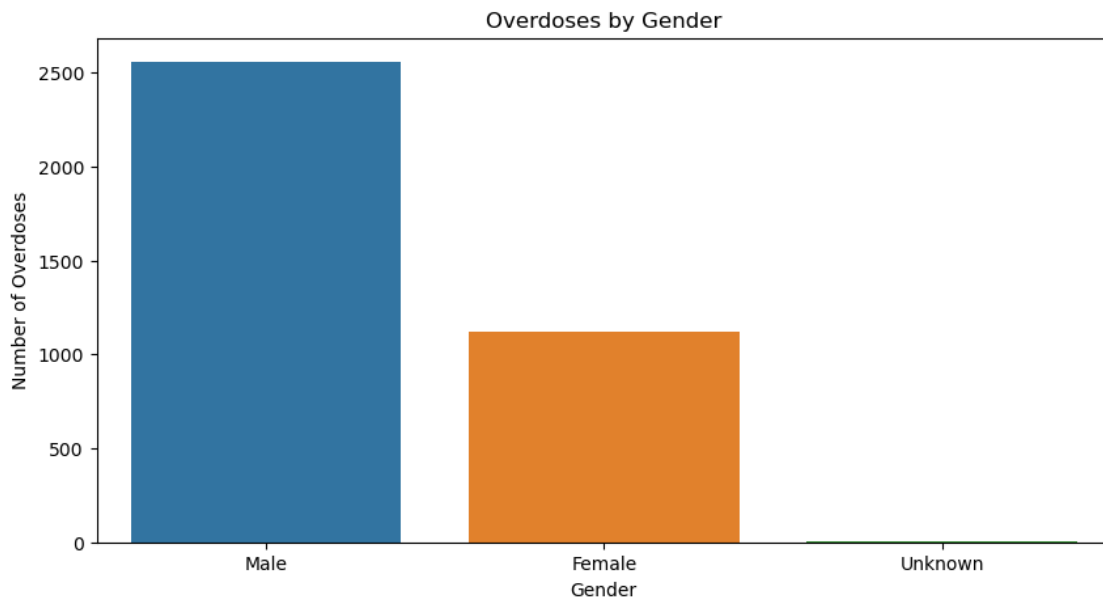
```

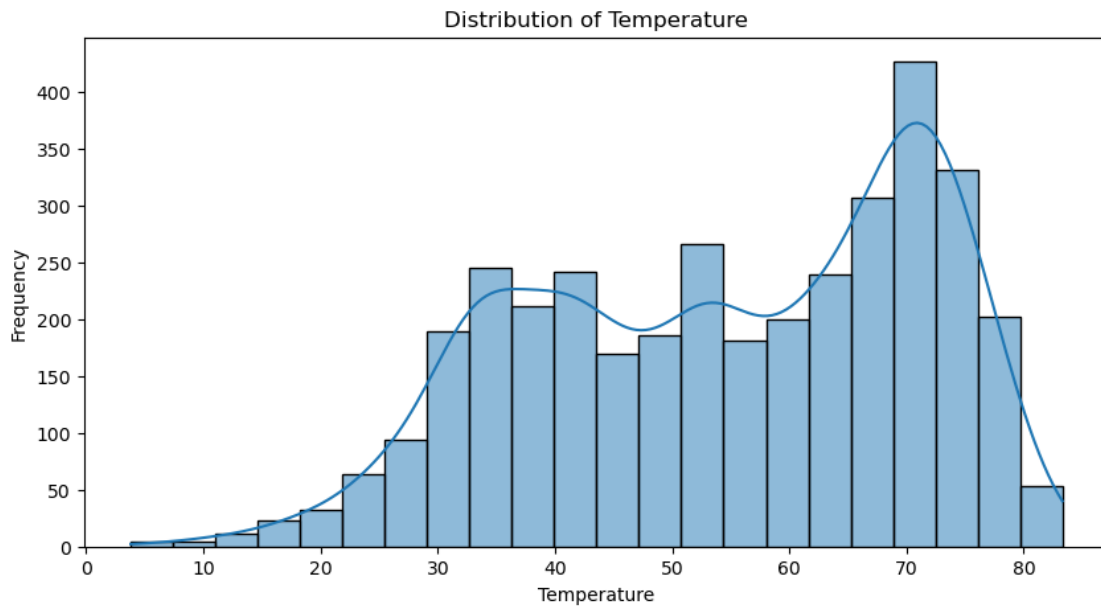
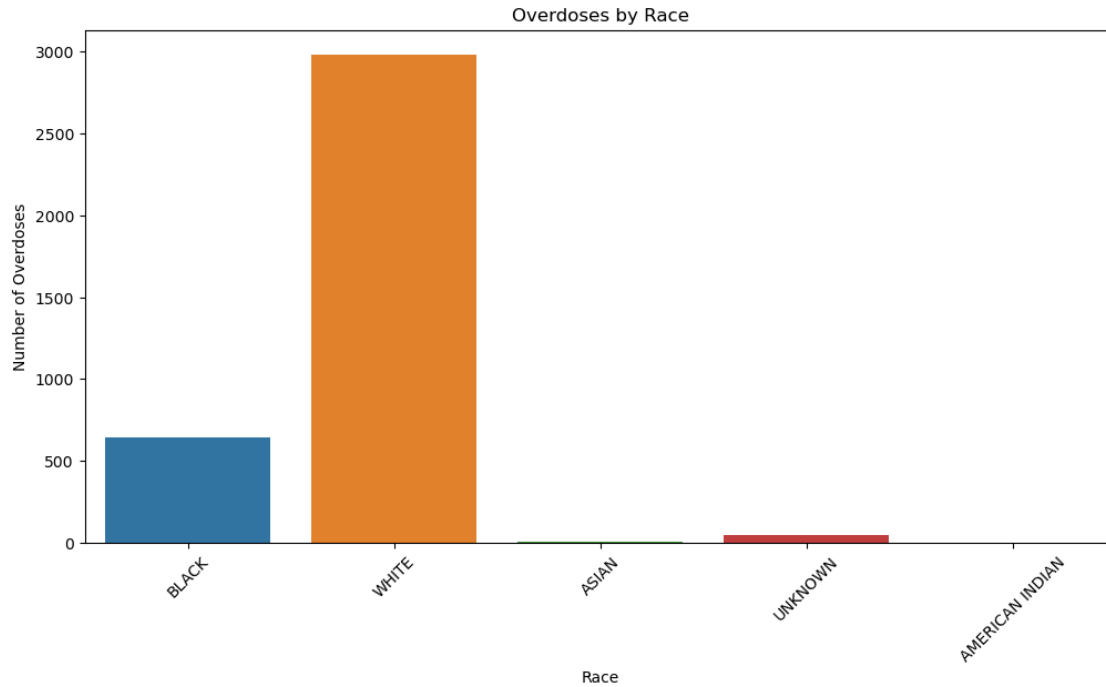
plt.xlabel('Race')
plt.ylabel('Number of Overdoses')
plt.xticks(rotation=45)
plt.show()

# Distribution of temperature values
plt.figure(figsize=(10, 5))
sns.histplot(df['temp'], kde=True)
plt.title('Distribution of Temperature')
plt.xlabel('Temperature')
plt.ylabel('Frequency')
plt.show()

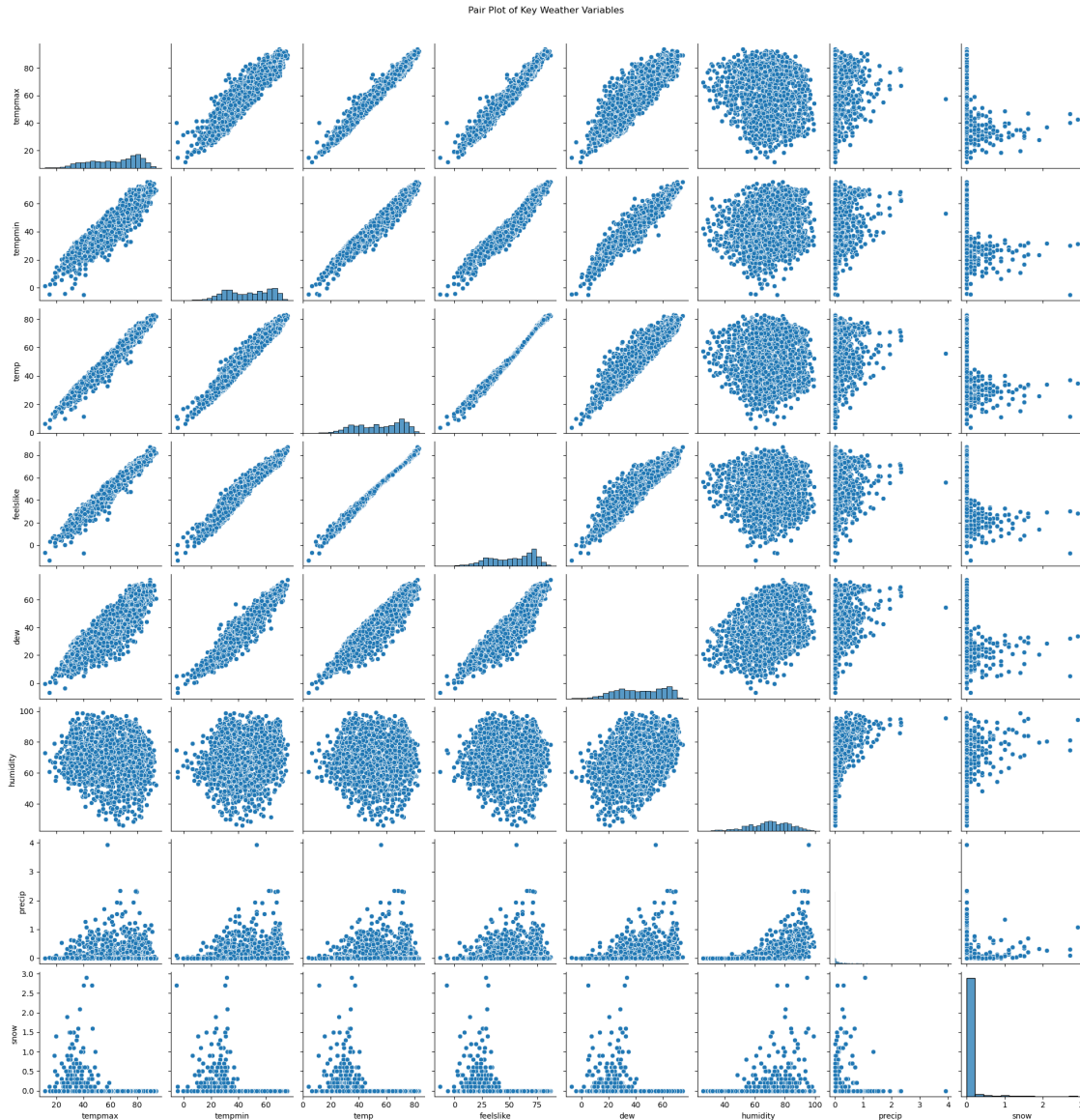
# Pair plot of key weather variables
sns.pairplot(df[['tempmax', 'tempmin', 'temp', 'feelslike', 'dew', 'humidity', 'precip', 'snow']])
plt.suptitle('Pair Plot of Key Weather Variables', y=1.02)
plt.show()

```





```
/Users/hunterberberich/anaconda3/lib/python3.11/site-  
packages/seaborn/axisgrid.py:118: UserWarning: The figure layout has changed to  
tight  
self._figure.tight_layout(*args, **kwargs)
```

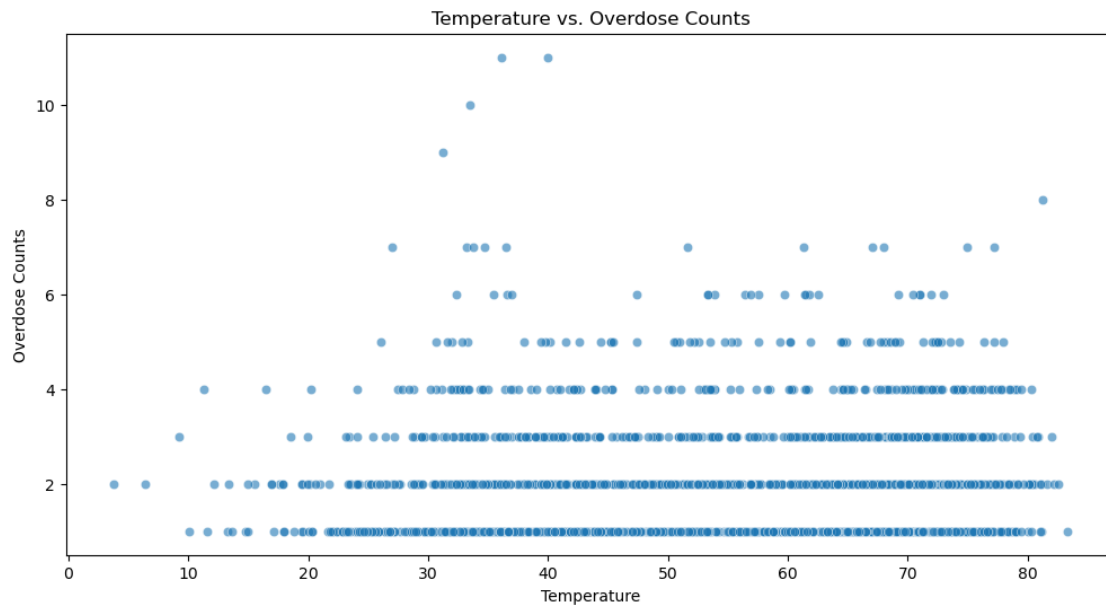


```
[ ]: df_aggregated = df.groupby('date').size().reset_index(name='overdose_count')

df_merged = pd.merge(df_aggregated, df[['date', 'temp']], on='date',
                    how='left').drop_duplicates()

# Scatter plot of temperature vs. overdose counts
plt.figure(figsize=(12, 6))
sns.scatterplot(data=df_merged, x='temp', y='overdose_count', alpha=0.6)
plt.title('Temperature vs. Overdose Counts')
plt.xlabel('Temperature')
```

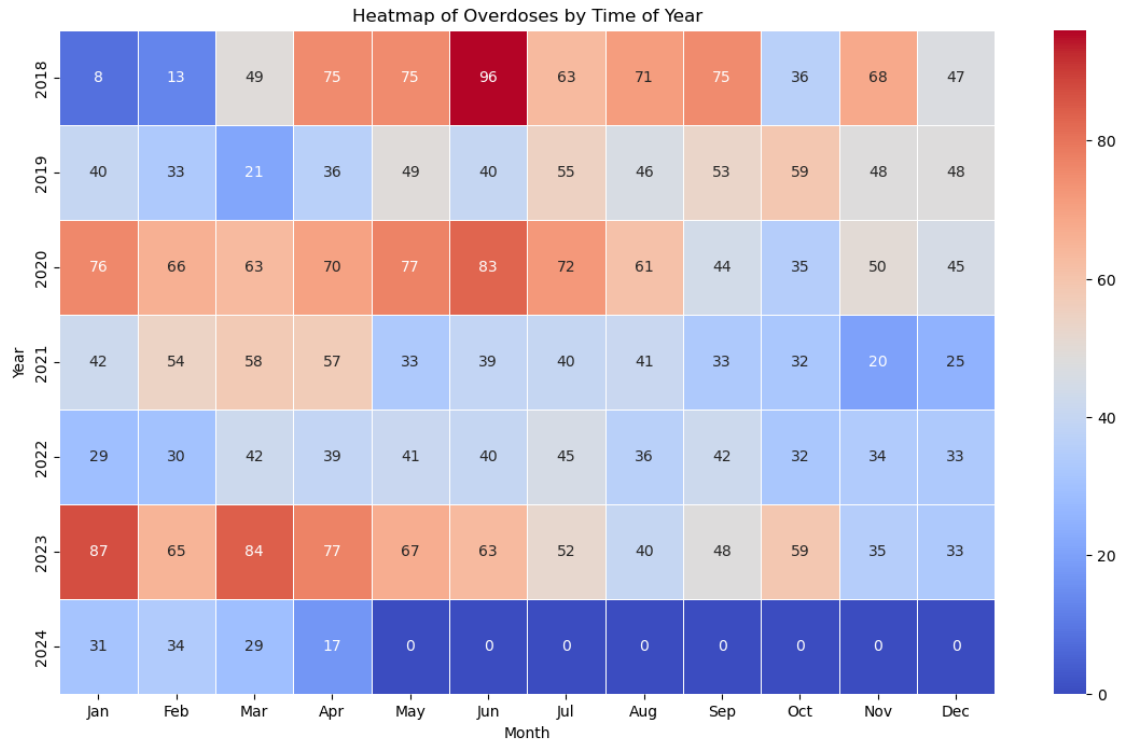
```
plt.ylabel('Overdose Counts')
plt.show()
```



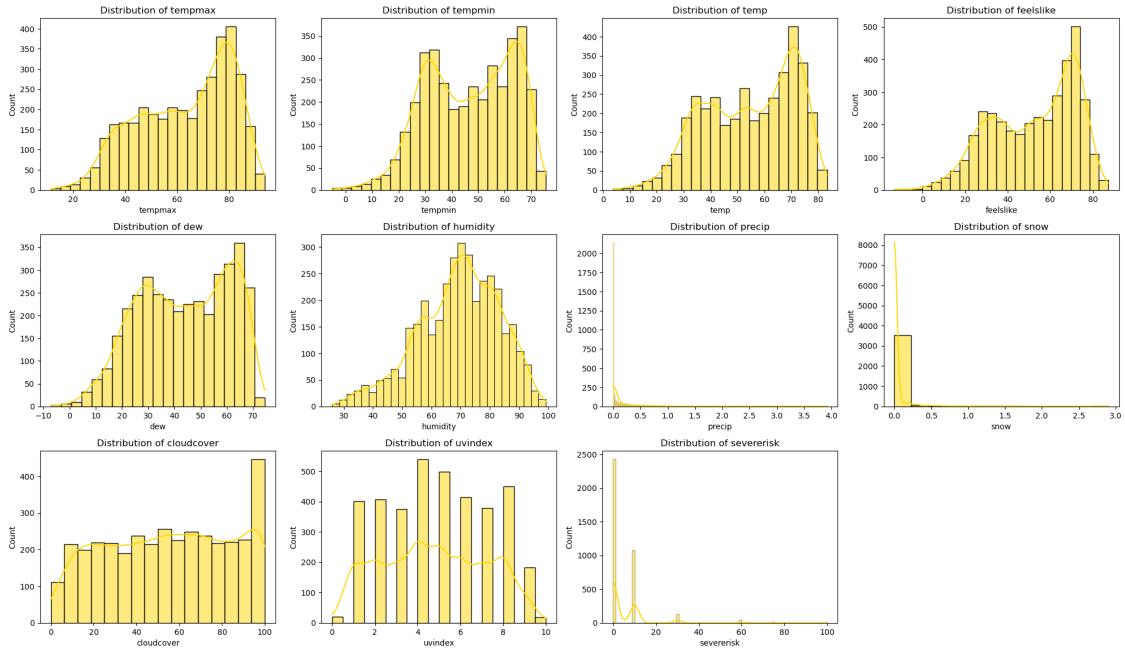
```
[ ]: monthly_overdoses = df.groupby(['year', 'month']).size().
    ↪reset_index(name='overdose_count')

monthly_overdoses_pivot = monthly_overdoses.pivot(index='year',
    ↪columns='month', values='overdose_count').fillna(0)
monthly_overdoses_pivot.columns = [calendar.month_abbr[m] for m in
    ↪monthly_overdoses_pivot.columns]

plt.figure(figsize=(14, 8))
sns.heatmap(monthly_overdoses_pivot, annot=True, fmt=".0f", cmap='coolwarm',
    ↪linewidths=0.5)
plt.title('Heatmap of Overdoses by Time of Year')
plt.xlabel('Month')
plt.ylabel('Year')
plt.show()
```



```
[ ]: # Select weather-related columns
weather_columns = ['tempmax', 'tempmin', 'temp', 'feelslike', 'dew',
                  'humidity', 'precip', 'snow', 'cloudcover', 'uvindex', 'severerisk']
plt.figure(figsize=(20, 15))
for i, column in enumerate(weather_columns, 1):
    plt.subplot(4, 4, i)
    sns.histplot(df[column], kde=True, color='#FFD700', edgecolor='black')
    plt.title(f'Distribution of {column}')
plt.tight_layout()
plt.show()
```



[]:

[]:

[]: sus